AM assignment Mod 6

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R Markdown

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When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✓ dplyr     1.1.4     ✓ readr     2.1.5
## ✓ forcats   1.0.0     ✓ stringr   1.5.1
## ✓ ggplot2   4.0.0     ✓ tibble    3.3.0
## ✓ lubridate 1.9.4     ✓ tidyr     1.3.1
## ✓ purrr     1.1.0
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
e errors
```

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.5.2
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(pROC)
```

```
## Warning: package 'pROC' was built under R version 4.5.2
```

```
## Type 'citation("pROC")' for a citation.
##
## Attaching package: 'pROC'
##
## The following objects are masked from 'package:stats':
##
##     cov, smooth, var
```

```
library(ggplot2)
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
library(class)
library(lpSolve)
```

```
## Warning: package 'lpSolve' was built under R version 4.5.2
```

Ojbective function:

A1, A2, A3 (plant A and warehouse 1,2,3), B1, B2, B3 (plant B and warehouse 1,2,3)

Objective: 622XA1 + 614xA2 + 630xA3 + 641xB1 + 645xB2 + 649xB3

Constraints:

Plant A capacity: xA1 + xA2 + xA3 ≤ 100

Plant B capacity: xB1 + xB2 + xB3 ≤ 120

AED Demand: XA1 + XB1 = 80, XA2 + XB2 = 60, XA3 + XB3 = 70

XAED ≥ 0

Name rows

```
plants <- c("Plant A", "Plant B")
warehouses <- c("WH1", "WH2", "WH3")
```

shipping and production cost, constraints per plant, demands per warehouse

```
shipcosts <- matrix(c(22, 14, 30, 16, 20, 24), nrow = 2, byrow = TRUE, dimnames = list(plants, w
arehouses))
product_costs <- c(600, 625)
plant_capacity <- c(100, 120)
plant_demand   <- c(80, 60, 70)
```

total costs for shipping and production for each unit

```
total_costs <- shipcosts + product_costs
total_costs
```

```
##         WH1 WH2 WH3
## Plant A 622 614 630
## Plant B 641 645 649
```

supply and demand constraints

```
objective_function <- lp.transport(cost.mat   = total_costs, direction  = "min", row.signs  = re
p("<=", 2), row.rhs = plant_capacity, col.signs  = rep("=",  3), col.rhs = plant_demand)
```

shipping solution

```
shipping <- objective_function$solution
dimnames(shipping) <- list(plants, warehouses)
```

print solution; AEDs totals at each plant

print solution; Plant capacity will reflect 10 unused AEDs remaining at plant B

print solution; two shipping options to plant A and B, including 10 unused AEDs to plant B reflect same

costs, meaning two optimal shipping options to plant A and plant B; cost to both = same $132,790.

```
cat("\nUnits produced: Plant A =", sum(shipping[1, ]),
    " Plant B =", sum(shipping[2, ]), "\n")
```

```
##
## Units produced: Plant A = 100  Plant B = 110
```

```
cat("Unused capacity: Plant A =", plant_capacity[1] - sum(shipping[1, ]),
    " Plant B =", plant_capacity[2] - sum(shipping[2, ]), "\n")
```

```
## Unused capacity: Plant A = 0  Plant B = 10
```

```
cat("Minimum total cost: $", format(objective_function$objval, big.mark = ","), "\n")
```

```
## Minimum total cost: $ 132,790
```